

Importing Libraries

```
In [40]: import pandas as pd
```

```
In [41]: import numpy as np
```

```
In [42]: import matplotlib.pyplot as plt
```

```
In [43]: import seaborn as sns
```

```
In [44]: %matplotlib inline
```

Loading Dataset

```
In [45]: epl_data=pd.read_csv(r"C:\Users\Bhawyl\Desktop\MINI_PROJECT\EPL_20_21.csv")
epl_data.head()
```

```
Out[45]:
```

	Name	Club	Nationality	Position	Age	Matches	Starts	Mins	Goals	Ass
0	Mason Mount	Chelsea	ENG	MF,FW	21	36	32	2890	6	
1	Edouard Mendy	Chelsea	SEN	GK	28	31	31	2745	0	
2	Timo Werner	Chelsea	GER	FW	24	35	29	2602	6	
3	Ben Chilwell	Chelsea	ENG	DF	23	27	27	2286	3	
4	Reece James	Chelsea	ENG	DF	20	32	25	2373	1	

```
In [46]: epl_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 532 entries, 0 to 531
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Name                                  532 non-null    object
1   Club                                  532 non-null    object
2   Nationality                           532 non-null    object
3   Position                              532 non-null    object
4   Age                                    532 non-null    int64
5   Matches                               532 non-null    int64
6   Starts                               532 non-null    int64
7   Mins                                  532 non-null    int64
8   Goals                                 532 non-null    int64
9   Assists                               532 non-null    int64
10  Passes_Attempted                      532 non-null    int64
11  Perc_Passes_Completed                 532 non-null    float64
12  Penalty_Goals                         532 non-null    int64
13  Penalty_Attempted                    532 non-null    int64
14  xG                                     532 non-null    float64
15  xA                                     532 non-null    float64
16  Yellow_Cards                          532 non-null    int64
17  Red_Cards                             532 non-null    int64
dtypes: float64(3), int64(11), object(4)
memory usage: 74.9+ KB
```

Counting the missing values

```
In [47]: epl_data.isna().sum()
```

```
Out[47]: Name                0
Club                0
Nationality         0
Position            0
Age                 0
Matches             0
Starts              0
Mins                0
Goals               0
Assists             0
Passes_Attempted    0
Perc_Passes_Completed 0
Penalty_Goals       0
Penalty_Attempted   0
xG                  0
xA                  0
Yellow_Cards        0
Red_Cards           0
dtype: int64
```

Add two new columns

```
In [48]: epl_data['MinsPerMatch']=(epl_data['Mins']/epl_data['Matches']).astype(int)
epl_data['GoalsPerMatch']=(epl_data['Goals']/epl_data['Matches']).astype(int)
epl_data.head()
```

Out[48]:

	Name	Club	Nationality	Position	Age	Matches	Starts	Mins	Goals	Ass
0	Mason Mount	Chelsea	ENG	MF,FW	21	36	32	2890	6	
1	Edouard Mendy	Chelsea	SEN	GK	28	31	31	2745	0	
2	Timo Werner	Chelsea	GER	FW	24	35	29	2602	6	
3	Ben Chilwell	Chelsea	ENG	DF	23	27	27	2286	3	
4	Reece James	Chelsea	ENG	DF	20	32	25	2373	1	

In [49]: *#total Goals*
 Total_Goals=epl_data['Goals'].sum()
 print(Total_Goals)

986

In [50]: Total_PenaltyGoals=epl_data['Penalty_Goals'].sum()
 print(Total_PenaltyGoals)

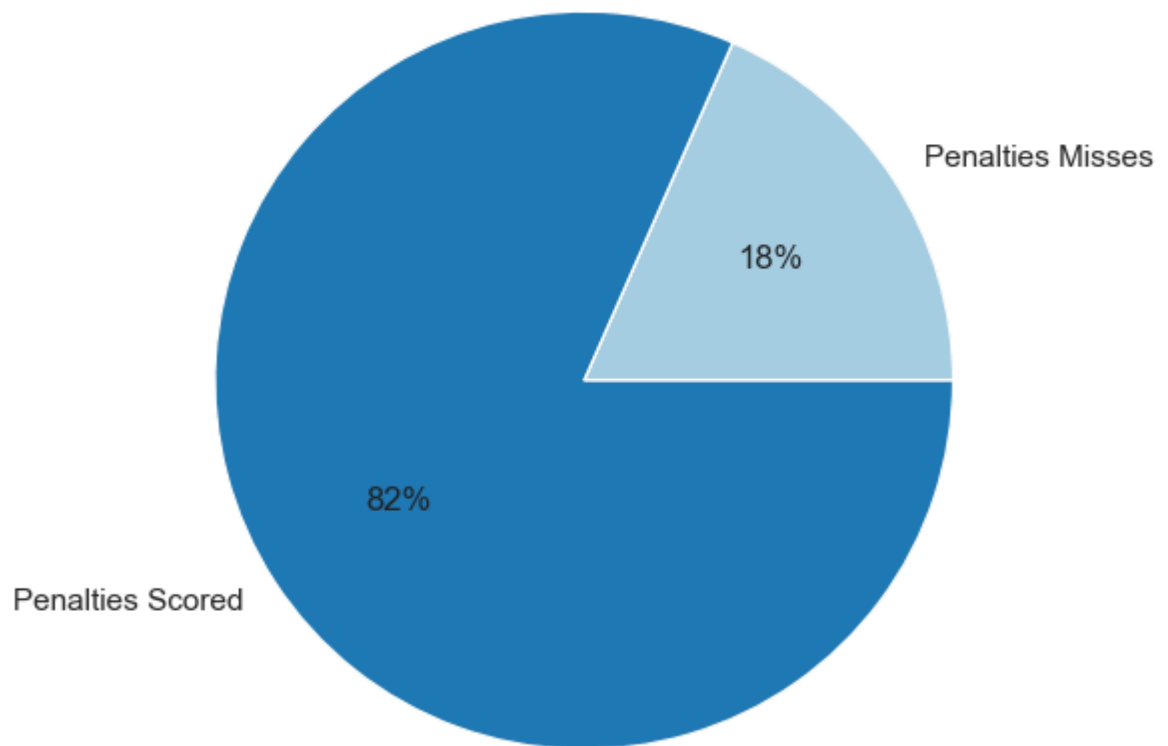
102

In [51]: Total_Penalty_Received=epl_data['Penalty_Attempted'].sum()
 print(Total_Penalty_Received)

125

Pie chart for penalties missed vs scored

In [52]: plt.figure(figsize = (13,6))
 penalty_not_scored = epl_data['Penalty_Attempted'].sum() - Total_PenaltyG
 data = [penalty_not_scored, Total_PenaltyGoals]
 labels = ['Penalties Misses', 'Penalties Scored']
 color_palette=sns.color_palette("Paired")
 plt.pie(data,labels=labels,colors=color_palette,autopct='%.0f%%')
 plt.show()



```
In [53]: epl_data['Position'].unique()
```

```
Out[53]: array(['MF,FW', 'GK', 'FW', 'DF', 'MF', 'FW,MF', 'FW,DF', 'DF,MF',  
               'MF,DF', 'DF,FW'], dtype=object)
```

```
In [54]: epl_data[epl_data['Position']=='FW']
```

Out[54]:

	Name	Club	Nationality	Position	Age	Matches	Starts	Mins	Goals
2	Timo Werner	Chelsea	GER	FW	24	35	29	2602	
16	Tammy Abraham	Chelsea	ENG	FW	22	22	12	1040	
19	Olivier Giroud	Chelsea	FRA	FW	33	17	8	748	
23	Ruben Loftus-Cheek	Chelsea	ENG	FW	24	1	1	60	
30	Raheem Sterling	Manchester City	ENG	FW	25	31	28	2536	
...	...	...	...	...	...	...	...	...	...
516	Oliver Burke	Sheffield United	SCO	FW	23	25	14	1269	
518	Oliver McBurnie	Sheffield United	SCO	FW	24	23	12	1324	
519	Rhian Brewster	Sheffield United	ENG	FW	20	27	12	1128	
523	Billy Sharp	Sheffield United	ENG	FW	34	16	7	735	
526	Daniel Jebbison	Sheffield United	ENG	FW	17	4	3	284	

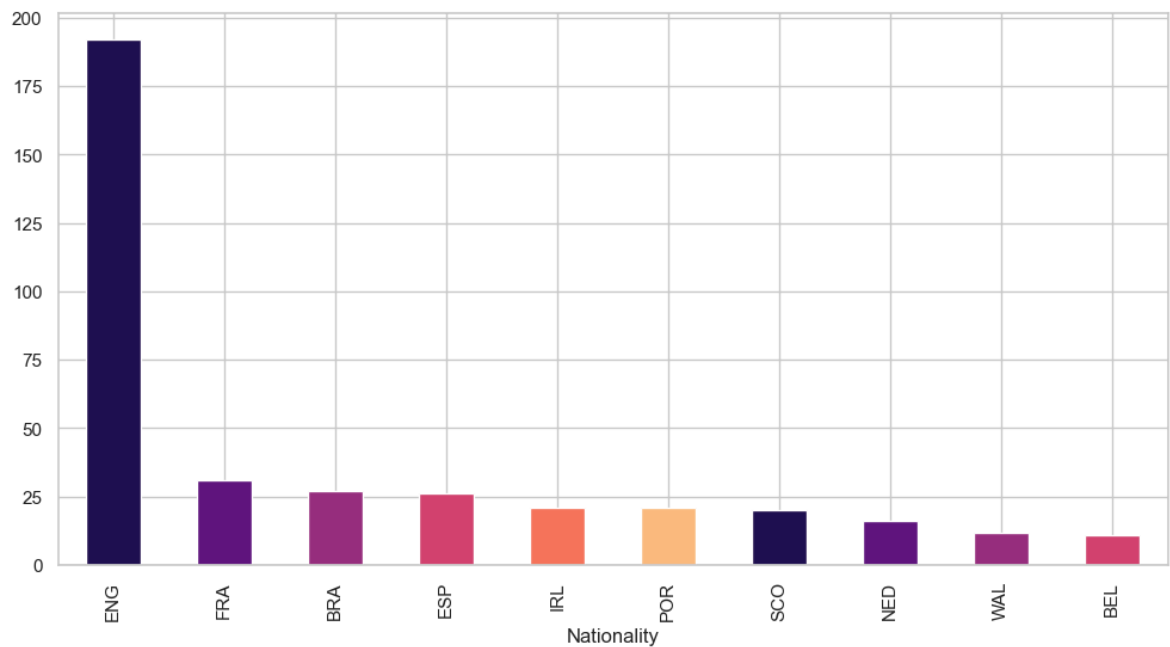
81 rows × 20 columns

```
In [55]: np.size((epl_data['Nationality'].unique()))
```

Out[55]: 59

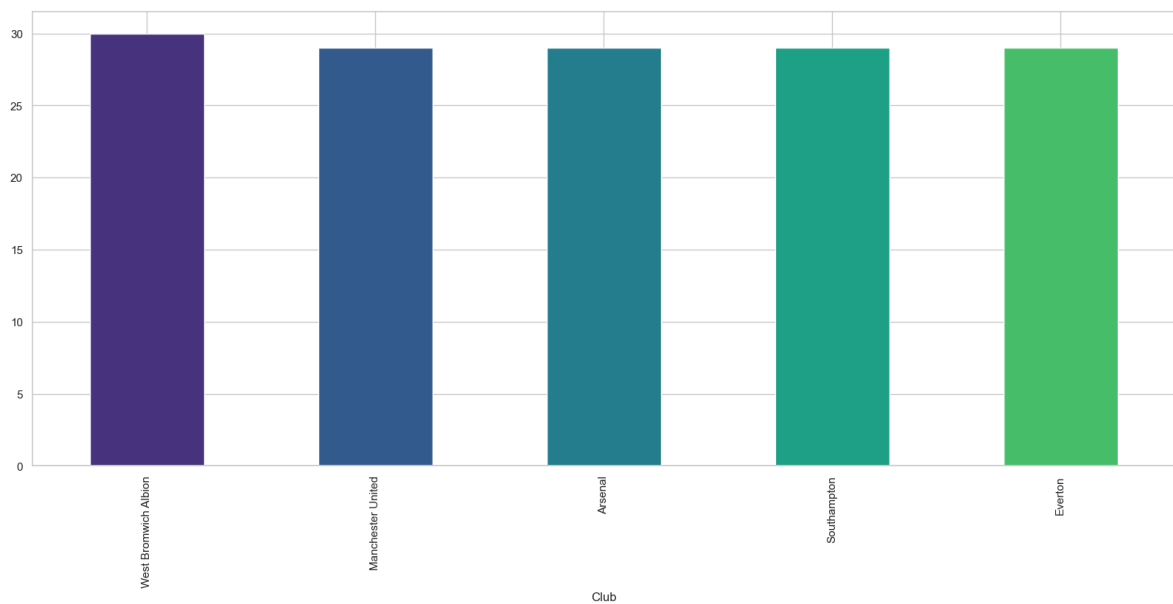
```
In [56]: nationality = epl_data.groupby('Nationality').size().sort_values(ascending=False)
nationality.head(10).plot(kind='bar', figsize=(12,6), color=sns.color_palette('magma'))
```

Out[56]: <Axes: xlabel='Nationality'>



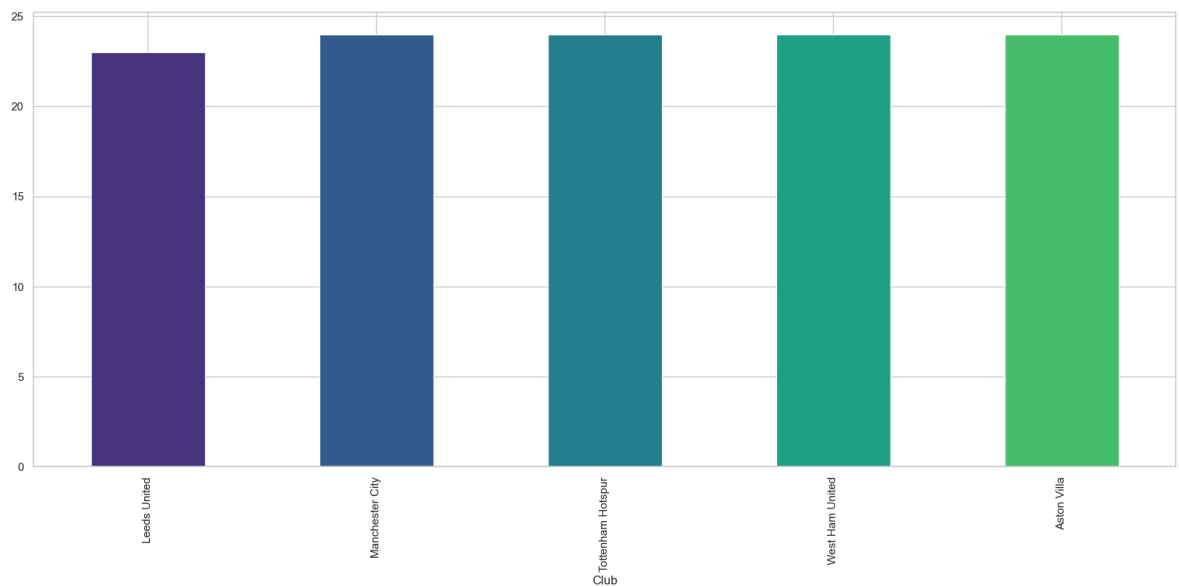
```
In [57]: #clubs with max players in their squad
epl_data['Club'].value_counts().nlargest(5).plot(kind='bar',color=sns.col
```

```
Out[57]: <Axes: xlabel='Club'>
```



```
In [58]: #clubs with latest players in their squad
epl_data['Club'].value_counts().nsmallest(5).plot(kind='bar',color=sns.co
```

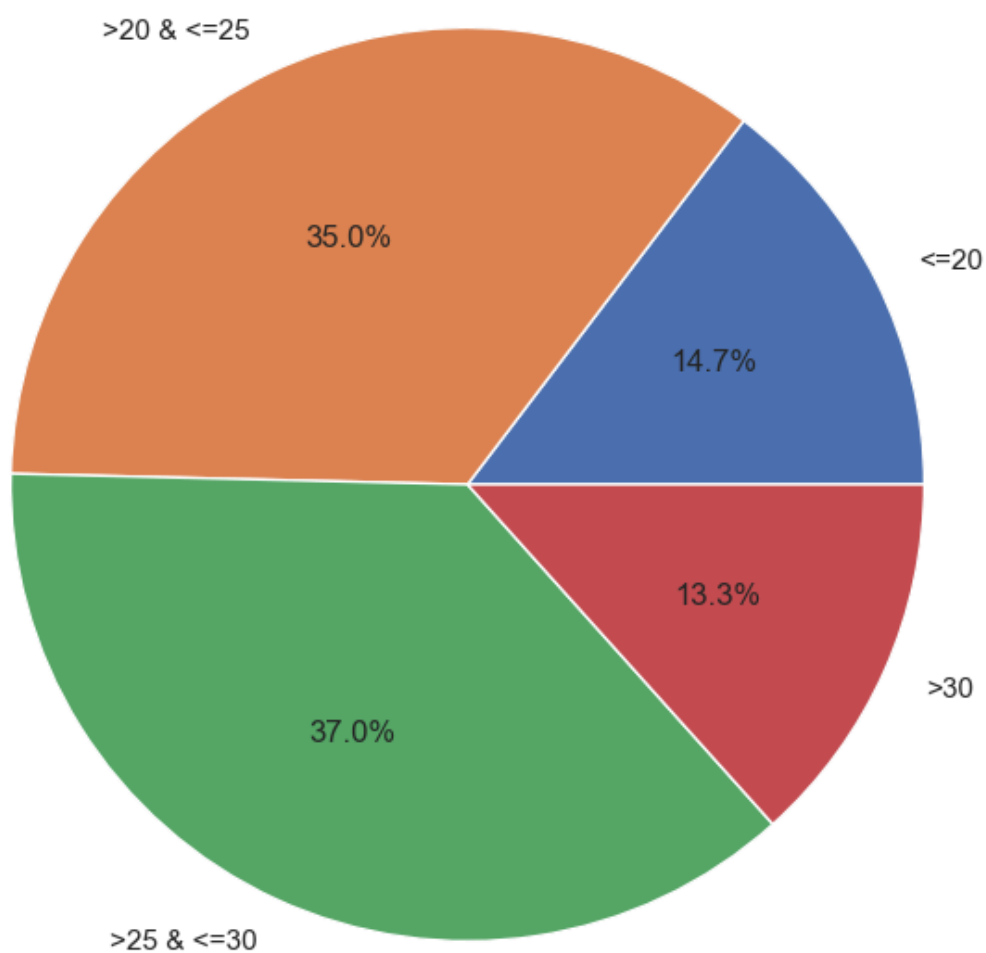
```
Out[58]: <Axes: xlabel='Club'>
```



```
In [59]: Under20 = epl_data[epl_data['Age']<=20]
Age20_25 = epl_data[(epl_data['Age']>20) & (epl_data['Age']<=25)]
Age25_30 = epl_data[(epl_data['Age']>25) & (epl_data['Age']<=30)]
Above30 = epl_data[epl_data['Age']>30]
```

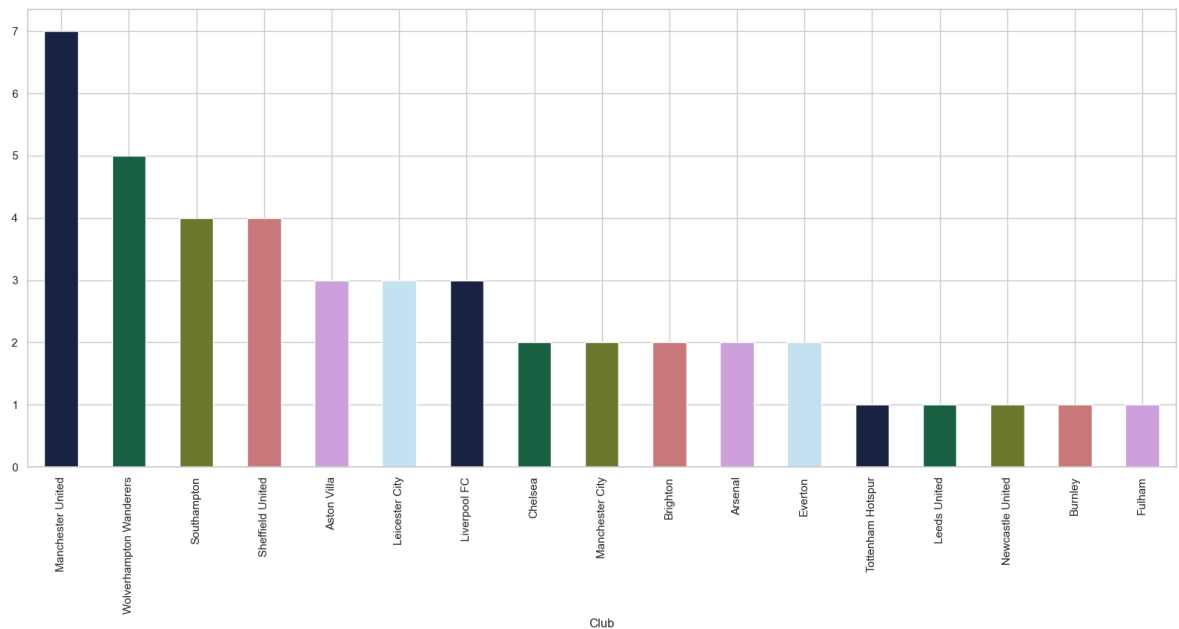
```
In [60]: x=np.array([Under20['Name'].count(),Age20_25['Name'].count(),Age25_30['Name'].count(),Above30['Name'].count()])
mylabels = ["<=20", ">20 & <=25", ">25 & <=30", ">30"]
plt.title('Total Players with Age Groups',fontsize=20)
plt.pie(x,labels=mylabels,autopct="%.1f%%")
plt.show()
```

Total Players with Age Groups



```
In [61]: players_under_20 = epl_data[epl_data['Age']<20]
         players_under_20['Club'].value_counts().plot(kind='bar', color=sns.color_
```

```
Out[61]: <Axes: xlabel='Club'>
```

```
In [62]: players_under_20[players_under_20["Club"]=="Manchester United"]
```

	Name	Club	Nationality	Position	Age	Matches	Starts	Mins	Goals
61	Mason Greenwood	Manchester United	ENG	FW	18	31	21	1822	
72	Brandon Williams	Manchester United	ENG	DF	19	4	2	188	
73	Amad Diallo	Manchester United	CIV	FW	18	3	2	166	
74	Anthony Elanga	Manchester United	SWE	FW	18	2	2	155	
76	Shola Shoretire	Manchester United	ENG	FW	16	2	0	11	
78	Hannibal Mejbri	Manchester United	FRA	MF	17	1	0	9	
79	William Thomas Fish	Manchester United	ENG	DF	17	1	0	1	

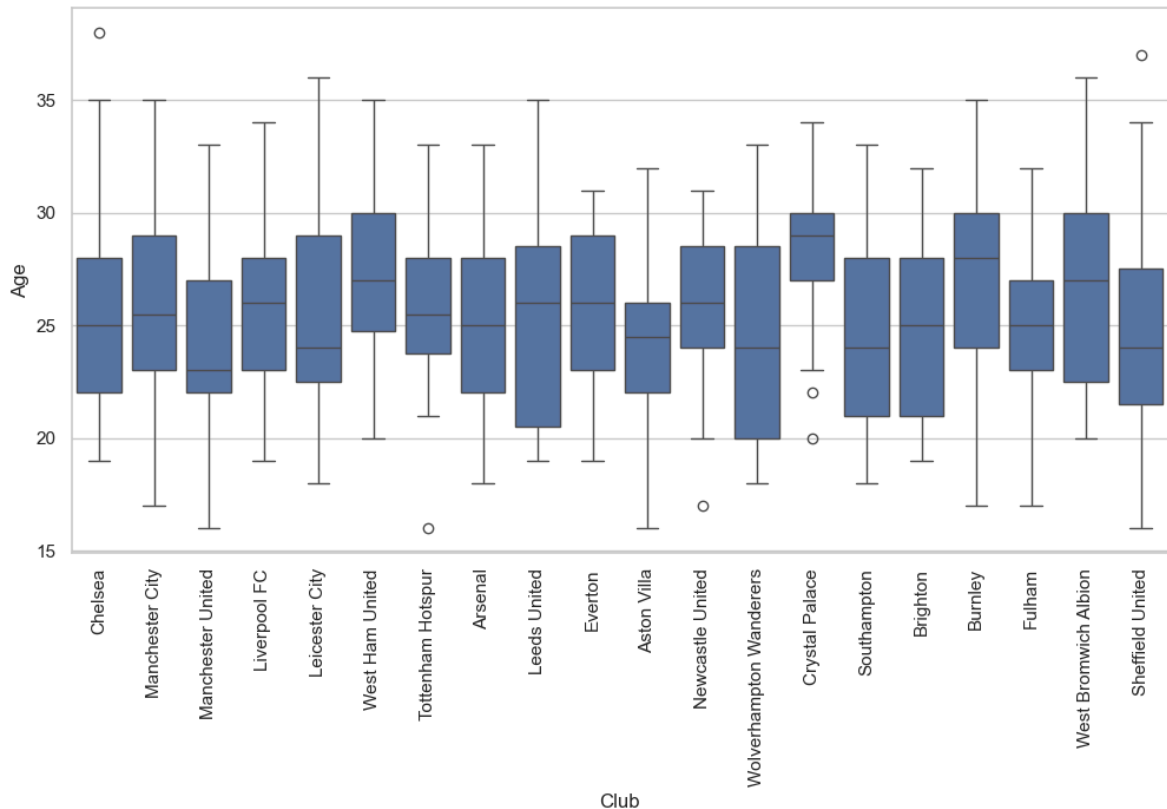
```
In [63]: players_under_20[players_under_20["Club"]=="Chelsea"]
```

	Name	Club	Nationality	Position	Age	Matches	Starts	Mins	Goals	Assists
18	Callum Hudson-Odoi	Chelsea	ENG	FW,DF	19	23	10	1059	2	
21	Billy Gilmour	Chelsea	SCO	MF	19	5	3	261	0	

```
In [64]: plt.figure(figsize=(12,6))
sns.boxplot(x='Club',y='Age',data=epl_data)
```

```
plt.xticks(rotation=90)
```

```
Out[64]: ([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
 [Text(0, 0, 'Chelsea'),
  Text(1, 0, 'Manchester City'),
  Text(2, 0, 'Manchester United'),
  Text(3, 0, 'Liverpool FC'),
  Text(4, 0, 'Leicester City'),
  Text(5, 0, 'West Ham United'),
  Text(6, 0, 'Tottenham Hotspur'),
  Text(7, 0, 'Arsenal'),
  Text(8, 0, 'Leeds United'),
  Text(9, 0, 'Everton'),
  Text(10, 0, 'Aston Villa'),
  Text(11, 0, 'Newcastle United'),
  Text(12, 0, 'Wolverhampton Wanderers'),
  Text(13, 0, 'Crystal Palace'),
  Text(14, 0, 'Southampton'),
  Text(15, 0, 'Brighton'),
  Text(16, 0, 'Burnley'),
  Text(17, 0, 'Fulham'),
  Text(18, 0, 'West Bromwich Albion'),
  Text(19, 0, 'Sheffield United')])
```



```
In [65]: num_player = epl_data.groupby('Club').size().sort_values(ascending=False)
print(num_player)
```

Club	
West Bromwich Albion	30
Arsenal	29
Everton	29
Manchester United	29
Southampton	29
Liverpool FC	28
Fulham	28
Brighton	27
Sheffield United	27
Wolverhampton Wanderers	27
Leicester City	27
Chelsea	27
Newcastle United	27
Burnley	25
Aston Villa	24
Crystal Palace	24
Tottenham Hotspur	24
Manchester City	24
West Ham United	24
Leeds United	23

dtype: int64

```
In [66]: epl_data.groupby('Club')['Age'].sum()
```

```
Out[66]: Club
Arsenal          724
Aston Villa      583
Brighton         663
Burnley          676
Chelsea          691
Crystal Palace   680
Everton          737
Fulham           701
Leeds United     583
Leicester City   691
Liverpool FC     716
Manchester City   617
Manchester United 692
Newcastle United 704
Sheffield United 670
Southampton      700
Tottenham Hotspur 615
West Bromwich Albion 803
West Ham United  660
Wolverhampton Wanderers 660
Name: Age, dtype: int64
```

```
In [67]: epl_data.groupby('Club')['Age'].sum()/num_player
```

```
Out[67]: Club
Arsenal 24.965517
Aston Villa 24.291667
Brighton 24.555556
Burnley 27.040000
Chelsea 25.592593
Crystal Palace 28.333333
Everton 25.413793
Fulham 25.035714
Leeds United 25.347826
Leicester City 25.592593
Liverpool FC 25.571429
Manchester City 25.708333
Manchester United 23.862069
Newcastle United 26.074074
Sheffield United 24.814815
Southampton 24.137931
Tottenham Hotspur 25.625000
West Bromwich Albion 26.766667
West Ham United 27.500000
Wolverhampton Wanderers 24.444444
dtype: float64
```

Total Assist from each club

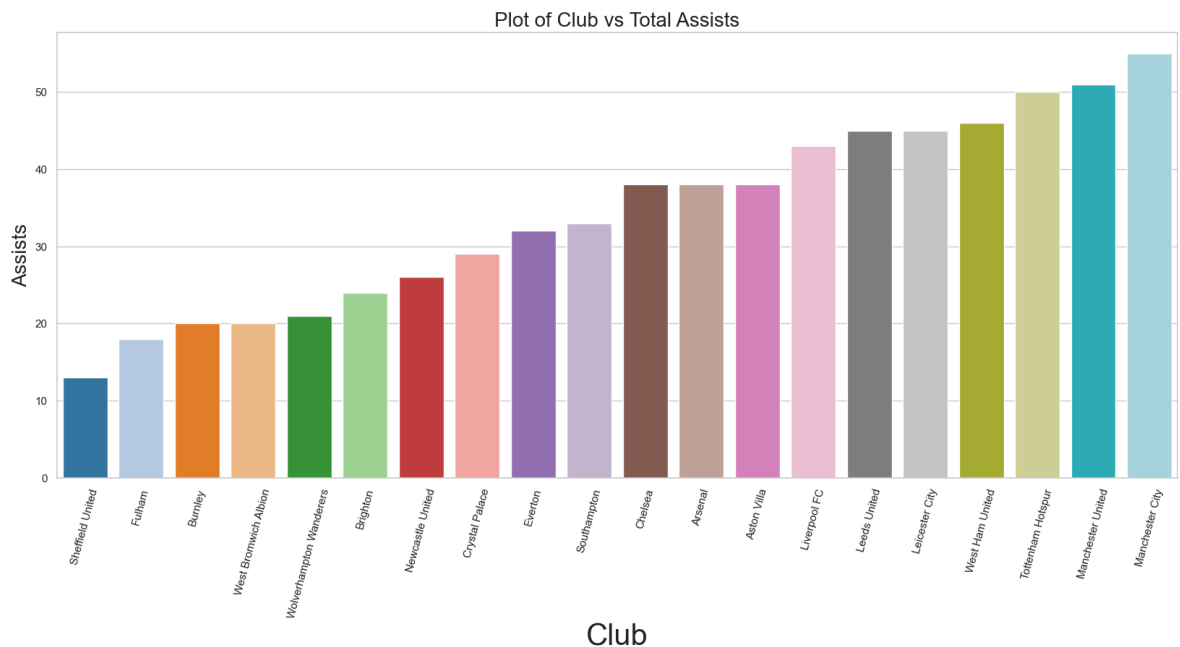
```
In [68]: Assists_by_club=pd.DataFrame/epl_data.groupby('Club', as_index=False)['Assists']
sns.set_theme(style='whitegrid',color_codes=True)
ax=sns.barplot(x='Club',y='Assists', data=Assists_by_club.sort_values(by='Assists'))
ax.set_xlabel("Club",fontsize=30)
ax.set_ylabel("Assists",fontsize=20)
plt.xticks(rotation=75)
plt.rcParams['figure.figsize']=(20,8)
plt.title('Plot of Club vs Total Assists',fontsize=20)
```

C:\Users\Bhawyl1\AppData\Local\Temp\ipykernel_3908\1120566133.py:3: Future Warning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
ax=sns.barplot(x='Club',y='Assists', data=Assists_by_club.sort_values(by='Assists'),palette='tab20')
```

```
Out[68]: Text(0.5, 1.0, 'Plot of Club vs Total Assists')
```



```
In [69]: top_10_assists=epl_data[['Name','Club','Assists','Matches']].nlargest(n=10)
top_10_assists
```

```
Out[69]:
```

	Name	Club	Assists	Matches
162	Harry Kane	Tottenham Hotspur	14	35
34	Kevin De Bruyne	Manchester City	12	25
51	Bruno Fernandes	Manchester United	12	37
161	Son Heung-min	Tottenham Hotspur	10	37
273	Jack Grealish	Aston Villa	10	26
54	Marcus Rashford	Manchester United	9	37
110	Jamie Vardy	Leicester City	9	34
220	Raphael Dias Belloli	Leeds United	9	30
2	Timo Werner	Chelsea	8	35
136	Aaron Cresswell	West Ham United	8	36

```
In [70]: # Creating a DataFrame to group the total goals scored by each club

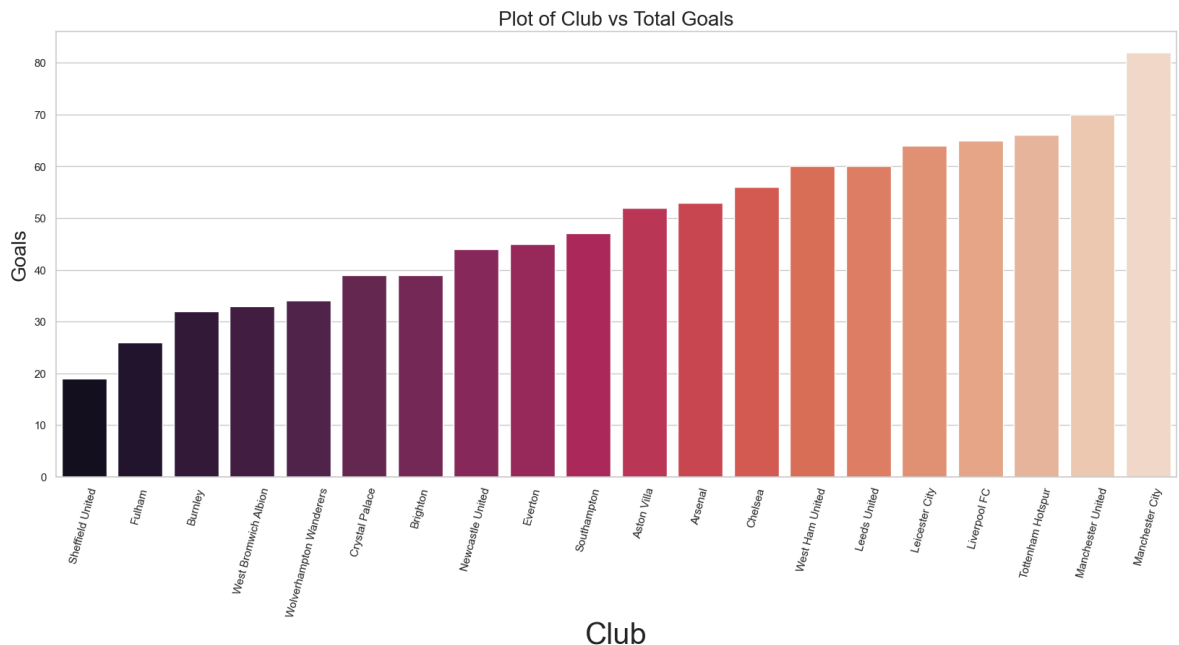
Goals_by_clubs = pd.DataFrame(epl_data.groupby('Club', as_index = False)[
sns.set_theme(style = "whitegrid", color_codes = True)
ax = sns.barplot(x = 'Club', y = 'Goals', data = Goals_by_clubs.sort_valu
ax.set_xlabel("Club", fontsize = 30)
ax.set_ylabel("Goals", fontsize = 20)
plt.xticks(rotation = 75)
plt.rcParams["figure.figsize"] = (20, 8)
plt.title('Plot of Club vs Total Goals', fontsize = 20)
```

```
C:\Users\Bhawyl1\AppData\Local\Temp\ipykernel_3908\588337942.py:5: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.
```

```
ax = sns.barplot(x = 'Club', y = 'Goals', data = Goals_by_clubs.sort_values(by = "Goals"), palette = 'rocket')
```

```
Out[70]: Text(0.5, 1.0, 'Plot of Club vs Total Goals')
```



```
In [71]: top_10_goals = epl_data[['Name', 'Club', 'Goals', 'Matches']].nlargest(n=top_10_goals)
```

```
Out[71]:
```

	Name	Club	Goals	Matches
162	Harry Kane	Tottenham Hotspur	23	35
81	Mohamed Salah	Liverpool FC	22	37
51	Bruno Fernandes	Manchester United	18	37
161	Son Heung-min	Tottenham Hotspur	17	37
214	Patrick Bamford	Leeds United	17	38
237	Dominic Calvert-Lewin	Everton	16	33
110	Jamie Vardy	Leicester City	15	34
267	Ollie Watkins	Aston Villa	14	37
33	İlkay Gündoğan	Manchester City	13	28
191	Alexandre Lacazette	Arsenal	13	31

```
In [72]: # Goals per match

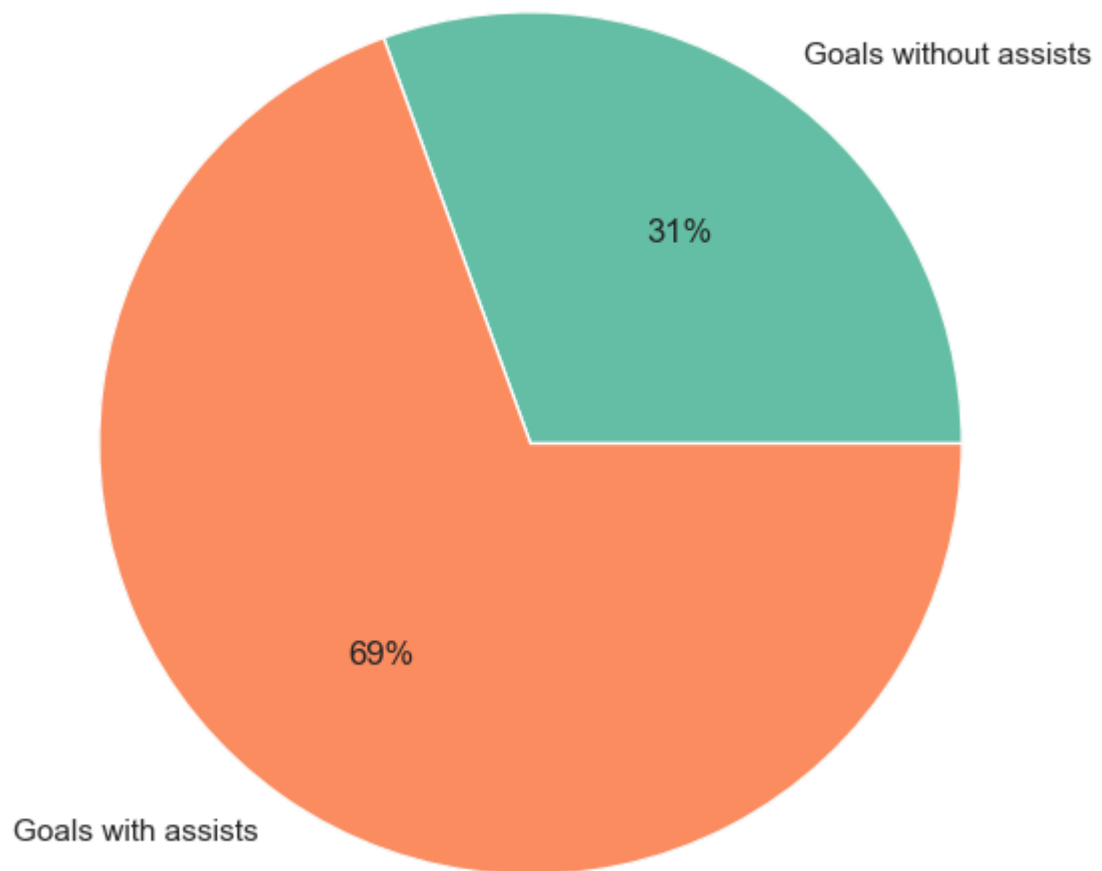
top_10_goals_per_match = epl_data[['Name', 'GoalsPerMatch', 'Matches', 'G
top_10_goals_per_match
```

Out[72]:

	Name	GoalsPerMatch	Matches	Goals
162	Harry Kane	0.657143	35	23
81	Mohamed Salah	0.594595	37	22
307	Joe Willock	0.571429	14	8
145	Jesse Lingard	0.562500	16	9
175	Gareth Bale	0.550000	20	11
74	Anthony Elanga	0.500000	2	1
51	Bruno Fernandes	0.486486	37	18
237	Dominic Calvert-Lewin	0.484848	33	16
120	Kelechi Iheanacho	0.480000	25	12
92	Diogo Jota	0.473684	19	9

```
In [73]: plt.figure(figsize = (14, 7))
assists = epl_data['Assists'].sum()
data = [Total_Goals - assists, assists]
labels = ['Goals without assists', 'Goals with assists']
color = sns.color_palette('Set2')
plt.pie(data, labels = labels, colors = color, autopct = '%.0f%%')
plt.title('Percentage of Goals with Assists', fontsize = 20)
plt.show()
```

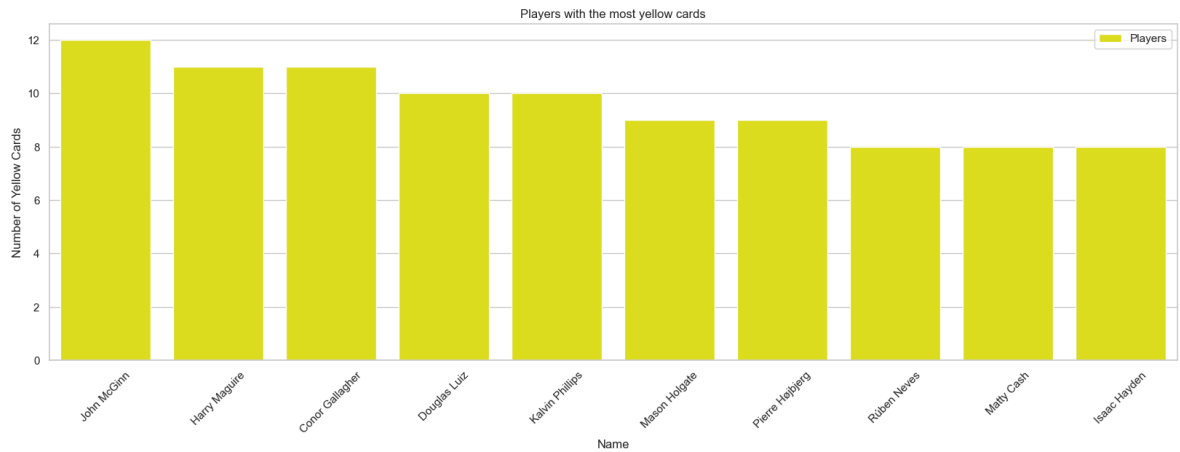
Percentage of Goals with Assists



```
In [74]: epl_yellow = epl_data.sort_values(by = 'Yellow_Cards', ascending = False)
plt.figure(figsize = (20, 6))
plt.title("Players with the most yellow cards")
c = sns.barplot(x = epl_yellow['Name'], y = epl_yellow['Yellow_Cards'], 1
plt.ylabel('Number of Yellow Cards')
c.set_xticklabels(c.get_xticklabels(), rotation = 45)
plt.show()
```

C:\Users\Bhawyl1\AppData\Local\Temp\ipykernel_3908\54215054.py:6: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i. e. after set_ticks() or using a FixedLocator.

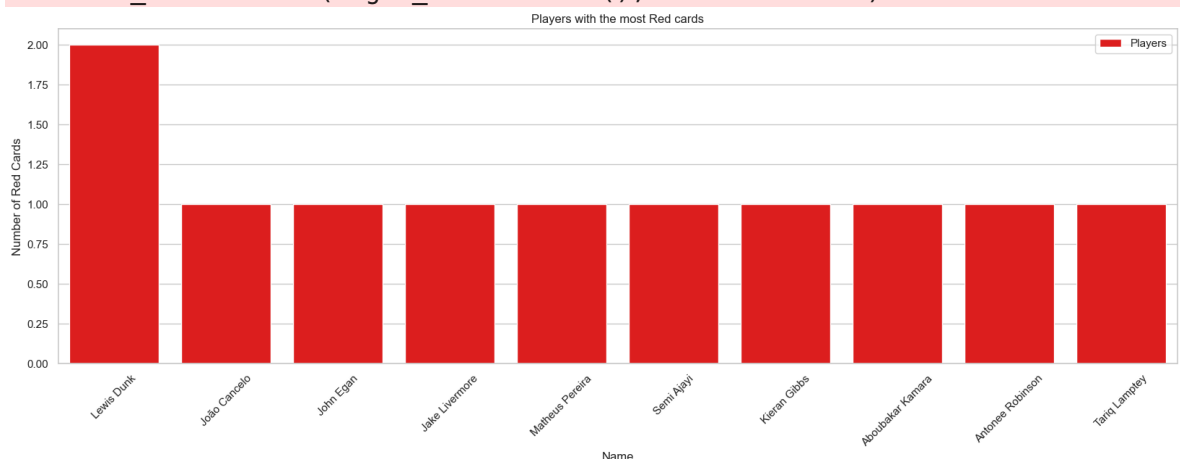
```
c.set_xticklabels(c.get_xticklabels(), rotation = 45)
```

```
In [75]: epl_yellow = epl_data.sort_values(by = 'Red_Cards', ascending = False)[:10]
plt.figure(figsize = (20, 6))
plt.title("Players with the most Red cards")
c = sns.barplot(x = epl_yellow['Name'], y = epl_yellow['Red_Cards'], label = 'Red_Cards')
plt.ylabel('Number of Red Cards')
c.set_xticklabels(c.get_xticklabels(), rotation = 45)
plt.show()
```

C:\Users\Bhawyl1\AppData\Local\Temp\ipykernel_3908\3731242498.py:6: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator.

```
c.set_xticklabels(c.get_xticklabels(), rotation = 45)
```



Performance Metrics

```
In [96]: # Create Goals per 90
epl_data['Goals_per_90'] = ((epl_data['Goals'] / epl_data['Mins']) * 90).

# Show table
table = epl_data[['Name', 'Club', 'Goals', 'Matches', 'Goals_per_90']]

print(table.head(10))
```

	Name	Club	Goals	Matches	Goals_per_90
0	Mason Mount	Chelsea	6	36	0.19
1	Edouard Mendy	Chelsea	0	31	0.00
2	Timo Werner	Chelsea	6	35	0.21
3	Ben Chilwell	Chelsea	3	27	0.12
4	Reece James	Chelsea	1	32	0.04
5	César Azpilicueta	Chelsea	1	26	0.04
6	N'Golo Kanté	Chelsea	0	30	0.00
7	Jorginho	Chelsea	7	28	0.31
8	Thiago Silva	Chelsea	2	23	0.09
9	Kurt Zouma	Chelsea	5	24	0.22

```
In [101... # assists per 90

epl_data['Assists_per_90'] = ((epl_data['Assists'] / epl_data['Mins']) *

table = epl_data[['Name', 'Club', 'Goals', 'Assists', 'Matches', 'Assists_per_
print(table.head(10))
```

	Name	Club	Goals	Assists	Matches	Assists_per_90
0	Mason Mount	Chelsea	6	5	36	0.16
1	Edouard Mendy	Chelsea	0	0	31	0.00
2	Timo Werner	Chelsea	6	8	35	0.28
3	Ben Chilwell	Chelsea	3	5	27	0.20
4	Reece James	Chelsea	1	2	32	0.08
5	César Azpilicueta	Chelsea	1	2	26	0.08
6	N'Golo Kanté	Chelsea	0	2	30	0.08
7	Jorginho	Chelsea	7	1	28	0.04
8	Thiago Silva	Chelsea	2	0	23	0.00
9	Kurt Zouma	Chelsea	5	0	24	0.00

```
In [111... epl_data['Goal_Contribution'] = epl_data['Goals'] + epl_data['Assists']
epl_data['Contribution_per_90'] = (epl_data['Goal_Contribution'] / epl_da

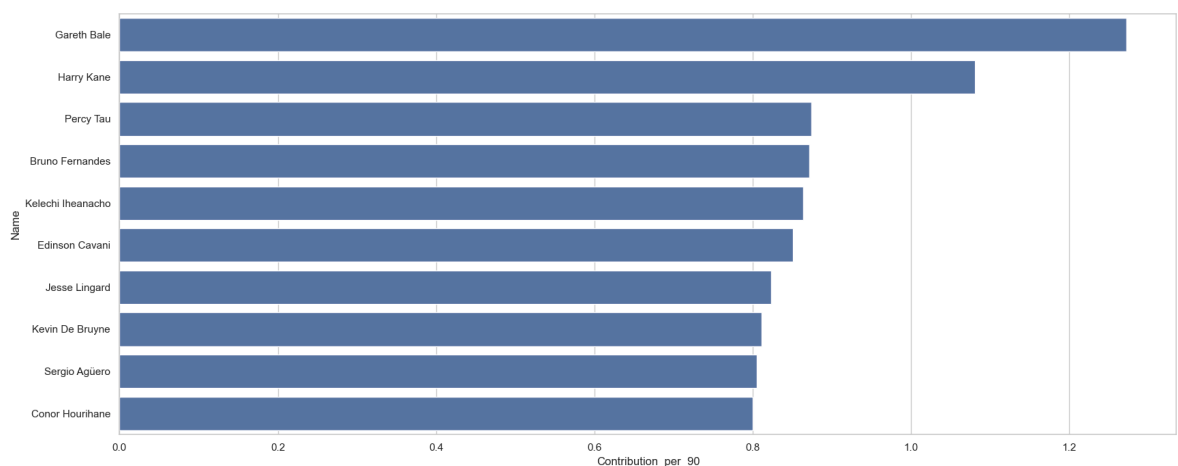
table1 = epl_data[['Name', 'Club', 'Goals', 'Assists', 'Matches', 'Contribution
table2 = epl_data[['Name', 'Club', 'Goals', 'Assists', 'Matches', 'Goal_Contri

print(table1.head(10))
print(table2.head(10))
```

	Name	Club	Goals	Assists	Matches	Contribution_per_9
0						
0	Mason Mount	Chelsea	6	5	36	0.3
4						
1	Edouard Mendy	Chelsea	0	0	31	0.0
0						
2	Timo Werner	Chelsea	6	8	35	0.4
8						
3	Ben Chilwell	Chelsea	3	5	27	0.3
1						
4	Reece James	Chelsea	1	2	32	0.1
1						
5	César Azpilicueta	Chelsea	1	2	26	0.1
2						
6	N'Golo Kanté	Chelsea	0	2	30	0.0
8						
7	Jorginho	Chelsea	7	1	28	0.3
6						
8	Thiago Silva	Chelsea	2	0	23	0.0
9						
9	Kurt Zouma	Chelsea	5	0	24	0.2
2						
	Name	Club	Goals	Assists	Matches	Goal_Contribution
0	Mason Mount	Chelsea	6	5	36	11
1	Edouard Mendy	Chelsea	0	0	31	0
2	Timo Werner	Chelsea	6	8	35	14
3	Ben Chilwell	Chelsea	3	5	27	8
4	Reece James	Chelsea	1	2	32	3
5	César Azpilicueta	Chelsea	1	2	26	3
6	N'Golo Kanté	Chelsea	0	2	30	2
7	Jorginho	Chelsea	7	1	28	8
8	Thiago Silva	Chelsea	2	0	23	2
9	Kurt Zouma	Chelsea	5	0	24	5

```
In [113... top = epl_data.sort_values('Contribution_per_90', ascending=False).head(1)
sns.barplot(x='Contribution_per_90', y='Name', data=top)
```

```
Out[113... <Axes: xlabel='Contribution_per_90', ylabel='Name'>
```

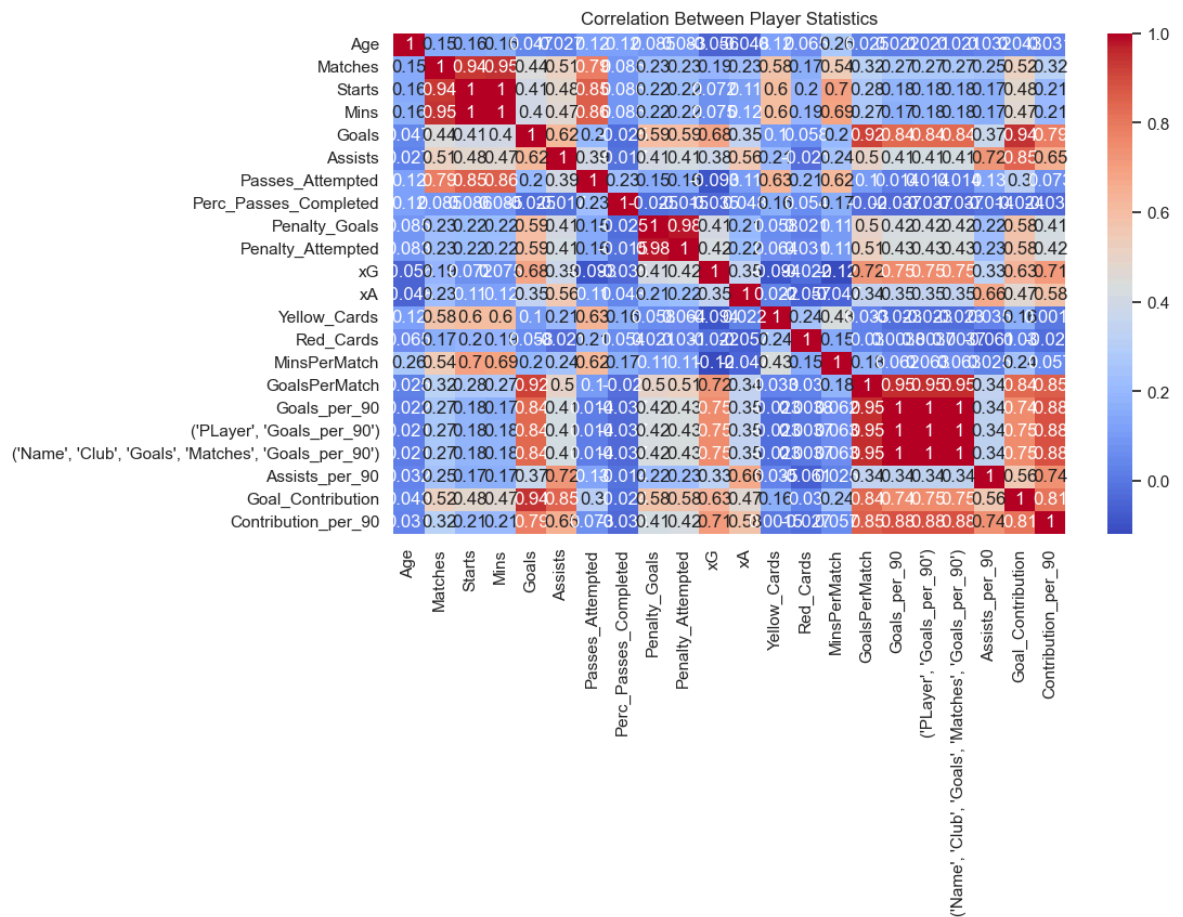


```
In [117... import seaborn as sns
import matplotlib.pyplot as plt

corr = epl_data.corr(numeric_only=True)

plt.figure(figsize=(10,6))
sns.heatmap(corr, annot=True, cmap='coolwarm')
```

```
plt.title("Correlation Between Player Statistics")
plt.show()
```



In []: